

TEACHING NOTE

SÉBASTIEN MARTIN

DRNY-2604-101

Transportation Crisis at Boston Metro School District

Case Synopsis

The Transportation Crisis at BMSD AI case is intentionally open-ended: Students interview five stakeholders role-played by AI (superintendent, CFO, COO, principal, and veteran driver) and, within the interaction, may obtain two shared resources (a district data workbook and a routing simulator) to build and defend their recommendations. Although the narrative is fictionalized, it draws closely on a 2017 collaboration between MIT researchers and Boston Public Schools (renamed Boston Metro School District in the case).

Students play the role of a consultant recruited by Superintendent Dr. Emily Carter to diagnose a transportation crisis at the Boston Metro School District (BMSD). They interview the five characters to understand how the district really runs, then propose a path forward across the choices the school district faces: how to use a new MIT routing optimizer; whether to stagger bell schedules; whether to close schools; and how to bring drivers, families, and the School Committee along.

The case was created for an operations management class but is well-suited to many curricula. Its most common use is to teach prescriptive analytics in practice, with the optimizer surfacing trade-offs that no static spreadsheet can. It also works for change management, the human-algorithm interface, public-sector operations, and the multi-objective tensions that arise when finance, equity, and union concerns all sit at the same table. Each character holds a different piece of the system; for instance, no understanding of the bus crisis can occur without talking to a driver. As a capstone, the case lets instructors pull together many threads of an MBA curriculum (analytics, operations, organizational behavior, leadership, stakeholder management) inside a single exercise.

The story the case tells is documented in two news articles referred to in later sections: *The Wall Street Journal*'s "How Do You Fix a School Bus Problem? Call MIT," which describes the early, successful use of the algorithm to redesign the district's bus routes; and *The Boston Globe*'s "Computers Can Solve Your Problem. You May Not Like the Answer," which details how the project later became a crisis when the district pushed the algorithm into more politically charged decisions.

Learning Objectives

After reading and analyzing the case, students will be able to:

- **Synthesize** information from multiple stakeholders with conflicting priorities and incomplete views to diagnose a complex operational and political problem
- **Apply** prescriptive analytics tools (optimization and what-if simulation) to a real decision and interpret their outputs critically, including the limits of aggregate metrics
- **Evaluate** the role of algorithms in public-sector decisions: when to trust them; what they hide; and how to govern their use across stakeholders
- **Design** a plan for the district to enact and defend it to skeptical stakeholders, integrating analytics, change management, equity, and political feasibility

Appropriate Uses

This case was designed and taught as a capstone for an operations management course to show how operations decisions unfold in practice; those decisions must comprise analytics, people, policy, and politics. However, the deliberately open-ended experience can be adapted to other contexts. The themes below are common choices; instructors should select and sequence them according to their courses.

- *Prescriptive optimization in the real world.*

The case provides a concrete entry point to prescriptive analytics. Students can identify decision variables (bus assignments, bell times); constraints (driver duty limits, capacity, legal windows); and objectives (minimize buses or ride-time). The simulator makes trade-offs visible: holding start times fixed; increasing riders per bus reduces fleet size but lengthens rides; staggering start times unlocks additional savings. A Pareto curve—buses versus mean ride-time—frames decisions as choices along a frontier rather than as a single solution.

- *Change management in operations.*

The characters embody the coalition-building work required to implement change: reconciling the CFO's urgency with the superintendent's politics, the driver's union realities, and the principal's concern regarding family impacts. Discussion can surface stakeholder sequencing, transparency, pilots, and the cost of losing trust.

- *Human-algorithm interface.*

The simulator is intentionally simple and fast, which makes it teachable while raising governance questions: How should leaders validate a black-box optimizer? Which metrics belong in public reports? How should uncertainties be communicated to families and staff? What division of labor between model and human planners yields the best outcomes?

- *Equity and policy.*

Transportation decisions distribute inconvenience and opportunity. Using the workbook and simulator together, students can test whether proposals improve or worsen fairness for

schools serving higher shares of disadvantaged students and discuss how race and socioeconomic status shape the politics of start-time changes. The *Globe* article anchors a candid conversation about why even equity-advancing plans can fail when the inconvenience falls on influential communities.

- *Operations from the ground up.*

The case rewards students for fully understanding how the district works. Many of the most consequential facts (driver workload, ridership realities, route timing constraints, equity gaps) come from interviews with the principal and the driver, not the data workbook. It models a broader lesson: operating any complex public organization starts with listening to the people who run it.

The case suits MBA, undergraduate, and executive-education audiences. MBAs treat it as a capstone that integrates analytics, organizational behavior, and stakeholder management; this was the original setting and remains the case's most common use today. Undergraduates use it as a grounded illustration of how textbook concepts (optimization, public policy, change management) play out in a messy real environment. Executive audiences engage with it as a realistic simulation of a public-sector crisis, with no specific prerequisites beyond general business literacy.

Supplementary Materials

The following may be used as supplemental material to accompany the case:

- Dimitris Bertsimas, Arthur Delarue, and Sébastien Martin, "Optimizing Schools' Start Time and Bus Routes," *Proceedings of the National Academy of Sciences* 116, no. 13 (March 2019): 5943–5948, <https://www.pnas.org/doi/10.1073/pnas.1811462116>.
- Jo Craven McGinty, "How Do You Fix a School-Bus Problem? Call MIT," *Wall Street Journal*, August 11, 2017, <https://www.wsj.com/articles/how-do-you-fix-a-school-bus-problem-call-mit-1502456400>.
- David Scharfenberg, "Computers Can Solve Your Problem. You May Not Like the Answer." *The Boston Globe*, September 21, 2018, <https://apps.bostonglobe.com/ideas/graphics/2018/09/equity-machine>.

Assignment Questions

This open-ended case can be deployed in many ways: as a deep multi-day team project, for example, or a shorter exercise that illustrates one or two specific themes. Because pre-class assignments depend heavily on the chosen format, I discuss them alongside the teaching plan in the Case Analysis section of this teaching note. As a baseline, I typically ask each student to interview every character individually before class and submit a one-paragraph summary of each character's perspective. Teams then write a memo to Superintendent Carter recommending a path forward, with the AI conversations attached as evidence. Instructors who want to keep the section lighter or remove it entirely can do so without loss. Everything is restated in context below.

Teaching Plan

The case discussion in my own MBA operations management course occupies a 90-minute session and is described in detail in the Case Analysis section. The case was designed to be flexible: Instructors may run it as a deep multi-day team project (my default); as a single live workshop; or as a hybrid of the two. The high-level structure of my version is summarized below; the section after this explains the choices and the reasoning behind them.

Activity	Instructional Strategy	Duration (mins)
Situating the crisis: what is happening in BMSD	Class discussion, mix of student synthesis and instructor narration of the real Boston routing story	15
Walking through each character	Class discussion guided by the instructor	20
The simulator and the trade-off	Pareto-curve walkthrough (Figure TN3)	10
Harvesting and debating recommendations	Team-by-team presentation followed by class debate	25
What happened in practice	Instructor-led narrative drawing on the <i>WSJ</i> and <i>Boston Globe</i> articles	20
Total		90

Many instructors run the case in shorter or different formats; see the Case Analysis section for alternatives.

Case Overview

Picture late summer in a big-city school district office. A five-person planning team hunches over routing software while overtime climbs and parents trade screenshots of 5:45 a.m. wake-ups. Down the hall, the superintendent prepares to brief the School Committee on a deficit so large that school closures are on the table if savings do not materialize; she also knows the longest rides fall disproportionately on disadvantaged students. The mayor's staff keeps asking whether transportation will blow the budget again, union leaders worry about duty limits and safety, and the CFO wants real savings this year; otherwise, consolidation becomes inevitable. The COO, trained in operations research, believes an MIT-style optimizer could cut dozens of buses and shorten planning cycles, but skepticism runs high among colleagues and community voices who have seen bold plans break on politics. Everyone is partly right, and no one can move alone.

That is the world students enter. Boston Metro School District (BMSD) carries \$150 million in annual transportation spending (12.5% of a \$1.2 billion budget) on top of a \$50 million deficit. Transportation outspends arts, athletics, and custodial budget lines combined, yet many disadvantaged students still face bus rides of more than 90 minutes. Driver shortages, aging fleets, and brittle politics create a classic public-operations bind: The math suggests efficiency gains; the lived reality complicates every step.

The story is inspired by Boston Public Schools’ 2017 collaboration with MIT’s Operations Research group, a project in which I participated. That work led to a paper in the *Proceedings of the National Academy of Sciences*, cited in the Supplementary Materials section of this teaching note, that explains the underlying optimization and anchors the simulator used in this case. Two companion readings add color from different angles: *The Wall Street Journal*’s “How Do You Fix a School Bus Problem? Call MIT” describes how algorithm-drafted routes saved money once planners layered in local judgment; and *The Boston Globe*’s “Computers Can Solve Your Problem. You May Not Like the Answer” shows how a unanimously approved start-time reform unraveled within two weeks amid protests. BMSD’s characters and school names are fictional, but the forces at play mirror those actual events.

This case matters because it lets instructors and students talk about optimization in the wild, the politics of “fairness” when winners and losers are visible, and the very human interface between algorithms and public institutions. It works as a capstone on prescriptive analytics, an operations change-management exercise, a human–algorithm conversation, and an equity-and-policy discussion—all covered in one story.

Student Experience

Students interview five AI-driven stakeholders (superintendent, CFO, COO, principal, veteran driver). The characters have distinct personalities and motivations, an aspect that helps draw students into the world presented by the case: The COO is excited about the optimizer and can actively guide students in using the simulator; the superintendent balances politics and equity; the CFO pushes for savings; the driver surfaces frontline realities and union concerns; and the principal centers on student experience.

This AI case is open-ended. No single PDF “contains the answer,” and students must ask solid questions, then follow up, and request the resources available in-role. Characters can share two items that make basic analysis possible without leaving the narrative: a district workbook (enrollment, disadvantaged share, bell times, baseline ride durations) and a Google Sheets–based MIT simulator (students set an average riders-per-bus target and see the implied buses and ride-time). Because the links align on school names, teams can merge insights across sources. This format adapts well to different levels of familiarity with US public schools; students who need context can ask for it, and those ready for analysis can proceed directly. In prior runs, instructors and students alike reported that the role-play deepened engagement and raised the quality of recommendations.

This format places real responsibility on instructors to set expectations with students about the depth of analysis they seek. A team running the case as a deep project can easily spend dozens of hours of combined work; a quick exploratory pass with one or two characters takes 30 minutes. Both are legitimate uses. I describe my own expectations in detail in the Case Analysis section.

Characters: Knowledge and Behavior

Boston Metro School District’s five AI characters all draw from the same underlying reality. They understand that the district was founded in 1647, that desegregation-era busing converted transportation into an equity instrument, and that the current crisis is driven by a transportation

budget that has ballooned to \$150 million (12.5% of a \$1.2 billion budget) after a 23% rise in three years. They know the district now runs 750 buses and drivers across 123 schools, each averaging 2.2 routes per morning and afternoon, with mean ride-times of 37 minutes but long tails past 90 minutes. Every character feels the political squeeze: a seven-member School Committee with four mayoral appointees; a mayor seeking re-election, who shuns controversy; vocal parent and community groups who mobilize quickly; skeptical media coverage; and union negotiations that could derail reforms. They also share awareness of the COO's partnership with MIT on an untested optimization algorithm, and they point students to the same supporting materials—the district data workbook for enrollment/start times/disadvantaged share, the simulator Google Sheet, and the PNAS research paper—whenever deeper analysis is needed. Interaction norms are equally consistent: characters treat the visitor as an outside consultant hired by Dr. Emily Carter; they stay fully in-role without acknowledging any AI framing; they explain jargon in plain English because they know the student lacks school-operations expertise; and they redirect questions beyond their remit to the stakeholder who owns that information. This section summarizes those shared baselines and character-specific nuances; instructors who need the verbatim prompts can reach out to me at sebastien.martin@kellogg.northwestern.edu.

Dr. Emily Carter, Superintendent

Emily Carter is a 52-year-old superintendent in her fifth year leading BMSD after a career as a principal and district administrator in Chicago and after earning a Harvard Graduate School of Education PhD. Carter is known internally for her equity-first instincts and for speaking in metaphor-rich stories about students and families, yet she now carries the scars of a failed 2023 start-time overhaul that consumed 18 months of engagement, triggered mass protests, cost three supportive School Committee members their seats, and drained her political capital. Carter understands the headline finances better than anyone: Transportation consumes \$150 million; the deficit is \$50 million; 10 fewer buses would save roughly \$2.1 million; and closing a school frees about \$8 million per year. She lays out the political chessboard for students, describing the seven-member School Committee (four mayoral appointees); the mayor's reluctance to back controversial moves ahead of re-election; and the readiness of community groups to oppose closures or algorithm-driven reroutes. Her current initiatives are a balancing act—considering one or two school closures on a short fuse; monitoring the MIT routing work while worrying about breakage; repairing strained relationships with the bus drivers' union; and bracing for the next round of labor talks. Carter expects the student's first conversation to cover the crisis narrative; the decisions she must make and by when; who controls which datasets; and any baseline questions about US public-school operations. She relies on regular check-ins with the mayor's education advisor (who urges caution), maintains a tenuous alliance with union leadership, and listens closely to principals split between pro-change advocates and risk-averse skeptics. When pressed beyond her expertise, she quickly redirects—detailed financial modeling belongs with CFO James Thompson, and technical deep dives on the algorithm belong with COO Sarah Lee—but she constantly challenges the student to synthesize across stakeholders and to propose actions that simultaneously respect equity, political feasibility, and fiscal reality.

Why talk to Carter: She anchors feasibility and politics—what can pass, by when, and at what cost—so teams avoid elegant but impossible plans.

Mr. James Thompson, Chief Financial Officer

James Thompson, age 48, arrived three years ago after 15 years in corporate finance and consulting and brings a Wharton MBA's hard-nosed demeanor. He is openly impatient, sometimes sardonic, and tends to dismiss operational caveats or equity debates as "soft." Thompson starts every conversation by hammering home the \$50 million deficit, insisting that transportation accounts for \$150 million of the \$1.2 billion budget, and reminding people that 750 buses and drivers equate to roughly \$200,000 each, including wages, fuel, maintenance, spare vehicles, and depots. He is constantly doing back-of-the-envelope math: removing 10 buses saves \$2.14 million; closing a school saves about \$8 million a year; and anything short of those magnitudes barely dents the problem. From Thompson's perspective, the district provides overly generous transportation service, the bus drivers' union already has favorable wages, and tinkering with MIT algorithms wastes time that could be spent cutting service to state minimums or consolidating schools. He controls the authoritative data workbook and withholds it until the student makes a precise, finance-relevant request. Expect him to advocate closing two or three schools, shrinking coverage, and celebrating any proposal that reduces bus count, even if it lengthens rides. Operational nuances or political optics are not his concern; he pushes those questions to Sarah Lee or Emily Carter.

Why talk to Thompson: He keeps teams quantitatively honest, forces savings math and feasibility bounds, and is often a student favorite; his brusque tone surprises many and pulls them into the case.

Ms. Sarah Lee, Chief Operations Officer

Sarah Lee is 45, holds an MIT master's degree in operations research, and has spent four years as COO after running operations for a national logistics company. She is meticulous, deeply technical, and unabashedly enthusiastic about optimization, yet she also empathizes with drivers and planners. Lee can brief students on the real routing workflow (stops, single-school routes, capacity, sequencing) and the practical limits that generate today's average rides near 37 minutes with long tails.

Lee led the collaboration with MIT and references the PNAS paper when explaining how the optimizer respects duty limits and targets a 10%–15% reduction in buses. She can guide students in using the Google Sheets simulator: setting up sensible experiments; interpreting buses-versus-ride-time trade-offs; testing start-time scenarios; and understanding when an input choice risks infeasible or undesirable outcomes. She does not inundate students with cell references or tab mechanics; instead, she coaches them through "what to try," "what a reasonable range looks like," and "how to read what the sheet is telling you."

From a change-management standpoint, Lee tends to move fast, favoring attrition over layoffs and framing algorithmic experimentation as inevitable. She downplays community politics and expects the student to show that the data support aggressive moves; when conversations turn to finance, she defers to Thompson, and when politics dominate, she yields to Carter or Principal Rodriguez.

Why talk to Lee: She translates the optimizer and routing process into concrete levers and constraints—and she actively helps students run the simulator and make sense of its outputs before proposing change.

Mr. David Rodriguez, Principal, Osprey Gardens K–8

David Rodriguez is a 42-year-old first-generation college graduate who earned his Ed.M. at Harvard, taught middle-school math for a decade, and has led Osprey Gardens—one of BMSD’s most economically disadvantaged schools—for six years. He still lives in the neighborhood, chairs community meetings, and sits on the Principals’ Equity Council, so he sees transportation through the lens of fairness. Rodriguez explains that Osprey Gardens starts at 7:15 a.m.; that many of his students endure 60- to 90-minute bus rides and thus wake up around 4:30 a.m.; and that affluent schools often secure later start times because their parents dominate public forums. He describes the toll on families juggling shift work or multiple jobs, on students who arrive exhausted, and on teachers who spend mornings re-energizing kids instead of teaching. When students bring up algorithms, he immediately worries about real-world friction: roads under construction, parents who miss the bus window, drivers who cannot reach narrow streets, or midyear route changes that sow chaos. He regularly cites research on later start times to argue for well-being over efficiency. Rodriguez uses these conversations to test the students’ values; he asks what guardrails they will adopt, how they plan to measure impact on disadvantaged students, and whether they have listened to families before recommending change. He does not pretend to know financial intricacies or engineering details and gladly points to Thompson or Lee for those answers, but he insists that any strategy quantify its equity implications and include mitigation for frontline disruption.

Why talk to Rodriguez: He centers on the lived experience of students and families and will insist that any plan include measurable equity guardrails.

Ms. Linda Johnson, Veteran Driver and Union Shop Steward

Linda Johnson, aged 52, has driven BMSD buses for 23 years, raised two children who graduated from the district, and lives in Dorchester. She currently runs the Osprey Gardens 7:15 a.m. route, followed by Beechwood at 8:30 a.m. and Kensington at 9:30 a.m., then reverses the loop each afternoon. As the union shop steward, she serves as the voice of drivers and opens every interaction by asserting that veteran status. Johnson paints a vivid portrait of daily life: the day begins at 5:45 a.m. with bus inspections; six runs leave almost no buffer, so any snowstorm, double-parked car, traffic accident, or late parent cascades into late pickups for the next school; radios and GPS units fail frequently; bus maintenance backlogs force drivers to keep marginal vehicles running; and solo drivers are responsible for up to 72 students without aides. She resents that planners require buses to stop at every location, even when only 40%–60% of assigned students ride, noting that the complete lack of real-time ridership tracking means routes are built on unrealistic 100% usage assumptions. Johnson’s union perspective is wary: earlier “efficiency” programs simply lengthened routes; senior drivers now fear algorithm-driven layoffs; younger drivers are exiting for other jobs with better pay and benefits; and health insurance erosion is a sore point. She is not anti-technology per se—she champions ridership-tracking and other tools that make the job safer—but she is skeptical of the MIT optimizer because every previous upgrade increased the workload without addressing its root causes. When student interviewers float strategic ideas, she reminds them to consider basic needs such as bathroom breaks, safe equipment, and realistic schedules, and she demands guarantees that any bus reductions happen through attrition rather than pink slips. Johnson does not offer budget math or high-level politics; instead, she narrates frontline realities, so the student understands exactly how executive decisions land on drivers and school students.

Why talk to Johnson: She is the source of on-the-ground truth—ridership, safety, and practicality—and often sparks low-cost, high-impact fixes others miss.

District Workbook and Simulator

Within the case world, characters routinely offer two shared artifacts. Understanding what each file contains helps instructors anticipate the analyses students might attempt and the equity questions they could raise.

District Information

The School District Data workbook (https://docs.google.com/spreadsheets/d/1Lh1up3CFTkr_69N50Vp-IIAxnNsQEzy5OxSTHJTO1A/copy) is a Google Sheet that CFO James Thompson and COO Sarah Lee routinely share in-character whenever students request district-level facts. Each row represents a school and includes the following:

- **School Name**
- **Grade Band**
- **Economically Disadvantaged Students:** percentage of enrolled students flagged as economically disadvantaged
- **School Start Time:** current bell time before any reforms
- **Enrollment:** total students served
- **Current Average Transportation Time:** mean minutes spent on the bus each morning/afternoon under current routes

These fields allow instructors (or advanced students) to slice the district by grade level, start-time clusters, or socioeconomic mix. Joining the disadvantaged share with the transportation-time column surfaces equity gaps—for example, identifying whether high-disadvantage schools endure the longest rides or whether start times concentrate hardship in specific neighborhoods.

MIT Simulator

COO Sarah Lee points students to a Schoolbus Simulator in Google Sheets (<https://docs.google.com/spreadsheets/d/12eTU61W4BJUHuW2ffsKH6znAxKTyaqTV0hyimpYNCOU/copy>). Its first tab (see **Figure TN1**) explains that yellow cells accept user input while green cells report results. Students supply a single control: the average number of children per bus (any value between 1 and 72, including fractions). Higher values drive efficiencies by packing buses, yet the sheet warns that feasibility issues or longer routes could emerge. The simulator then reports two aggregated outputs: (1) estimated buses/drivers required and (2) the resulting district-wide average ride duration.

Figure TN1: Simulator Control Panel (Tab 1)

Input:	To guide the bus routing optimization algorithm, you must enter "how full" you want the buses to be <i>on average</i> (between 1 and 72 children per bus, the number can be fractional). However, it may not always be feasible for the algorithm to find a solution given the district constraints. Higher numbers lead to fewer buses needed overall, but the average length of the bus routes increases.	26	children per bus
Simulator Outputs:	Estimated number of buses (and bus drivers) needed to transport the children of the district to and from school everyday.	822	Buses
	Average time children have to spend on the bus to get to school every morning (and to come back home every afternoon)	0:26:22	
<i>You can also check a school-by-school analysis of the simulator's output in the "Schools" tab</i>			

Note: The yellow input sets the average riders per bus; the green outputs display the number of buses required and the mean ride-time.

The second tab (see **Figure TN2**) lists each school with editable start times, the number of buses assigned, and the average ride-time.

Figure TN2: School-Level View with Editable Start Times and Simulated Bus Demand by Campus (Tab 2)

School Name	School Start Time (can be modified if needed)	Number of bus used every morning/afternoon	Average time children spend in the bus every morning/afternoon
Adlark ES	7:30:00 AM	10	0:21:09
Alton Elementary	7:30:00 AM	4	0:13:12
Anthem Course To College	8:00:00 AM	9	0:21:54
Balwin ELC	7:30:00 AM	6	0:25:04
Baxter ES	9:30:00 AM	8	0:18:26
Beacon Comm Lead Acad	7:55:00 AM	14	0:39:58
Beechwood ES	8:30:00 AM	11	0:32:26
Blakely ES	8:30:00 AM	17	0:33:09
Bridgeport Arts Academy	8:00:00 AM	13	0:20:31
Brookline DAY AND EVENING ACADEMY	9:00:00 AM	14	0:17:23
Brighton Green Acad	8:00:00 AM	16	0:24:35
Bridgeway INTERNATIONAL HIGH	8:00:00 AM	10	0:18:32
Brookside Latin Academy	7:20:00 AM	45	0:18:48
Briarwood Latin School	7:45:00 AM	85	0:11:30
Bridgewater Teachers Union K-8	8:30:00 AM	11	0:23:09
Brampton ES	9:30:00 AM	9	0:17:20
Brockton HS	7:30:00 AM	23	0:36:02
Burnside HS	7:25:00 AM	16	0:25:31
Carrington	9:30:00 AM	1	0:44:50
Chatham Elementary	8:30:00 AM	8	0:36:49
Charleston High School	7:30:00 AM	28	0:40:08
Chiswick ES	8:30:00 AM	8	0:24:25
Clayton ES	9:30:00 AM	6	0:17:30
Coastal Acad Sci & Health	7:30:00 AM	11	0:34:42
Cornerstone ACADEMY	8:00:00 AM	2	0:29:05

Experiment levers:

- **Target riders per bus:** Increasing the target reduces the modeled bus count but raises the average time students spend riding. Lowering the target reverses the trade-off.

- **Start-time edits:** Because the model assumes buses can sequence multiple runs when bell times are staggered, spreading schools across the three bell slots lowers the total number of buses. Clustering many schools at the same bell time forces more buses and drivers.
- **Advanced what-ifs:** Instructors can demonstrate how a single school's shift ripples through system demand by editing its start time and observing adjustments in the per-school bus count column.

The simulator's per-school view aligns with the district workbook naming, so teams can merge the two sources to compare "before" vs. "after" ride-times, bus allocations, or start-time proposals. Instructors can encourage students to quantify not just the mean change but also which schools (or student populations) experience the largest swings.

Case Analysis

This section describes how I typically run the case in my MBA operations management course at the Kellogg School of Management, where I assign it near the end of the quarter as a capstone project. The case was developed to be flexible, and other instructors run it in shorter, lighter, or different formats. Treat this as one well-traveled path, not a prescription.

Pre-Class Work

Students do two pieces of work before the class session:

- **Individual exploration.** Each student interviews every character on their own and submits a short paragraph for each one summarizing what the character knows, what they care about, and what the student took away from the exchange. This pushes students to listen carefully to all five voices rather than coast on a teammate's notes. In my class, I budget about an hour of individual work per student. Strong students often have up to 100 messages per character; the case rewards depth of engagement.
- **Team memo.** After the individual round, teams of about five reconvene to draft a one- or two-page memo to Superintendent Dr. Emily Carter that recommends a path forward and explains the reasoning. Memos must be evidence-based; every claim should trace to a character conversation or a document a character provided. Students submit both the memo and the AI conversations they relied on so that I can verify sourcing and grade for analytic depth, not polish. I budget about five hours of teamwork in my class, but the actual range is wide.

The case also works with much lighter pre-work. If the goal is a one-hour illustration of a single theme rather than a deep analytical exercise, instructors could ask students to interview only one or two characters and skip the memo.

Format Choices

The format I described above is one option; here are some alternatives I have seen work:

- **Deep team project.** The default in my class: multi-day pre-work, a team memo, and a live debrief session. Best for capstone use.
- **Live workshop.** A single session in which students do their first interviews in class (in breakout rooms or at a shared screen), capture insights on a whiteboard, and co-create recommendations live. Best for a shorter exposure with strong collaborative energy.
- **Hybrid.** Require a minimum number of interviews before class, then use the live session to pressure-test numbers, map stakeholder coalitions, or design implementation roadmaps.

The right format depends on how much class time instructors have, how much pre-work they demand, and which themes they want to emphasize.

The Class Session

In my course, the live debrief is a 90-minute session organized in three segments: a collective analysis of what students learned (about 45 minutes); an open debate of recommendations (about 25 minutes); and a closing narrative of what really happened in Boston (about 20 minutes). The proportions can shift, but I have found this ordering important; students need a shared picture of the situation before they argue solutions, and they need to have argued solutions before the real story lands.

1. Situating the crisis and character debrief [~45 Minutes]

I open by asking the room to explain: “What is happening in this district?” The question forces students to synthesize everything the interviews revealed: the \$150 million transportation bill within a \$1.2 billion budget; the \$50 million deficit that threatens school closures; the driver shortages; and the MIT routing simulator that promises relief. I deliberately keep the class in this synthesis mode rather than jumping to solutions.

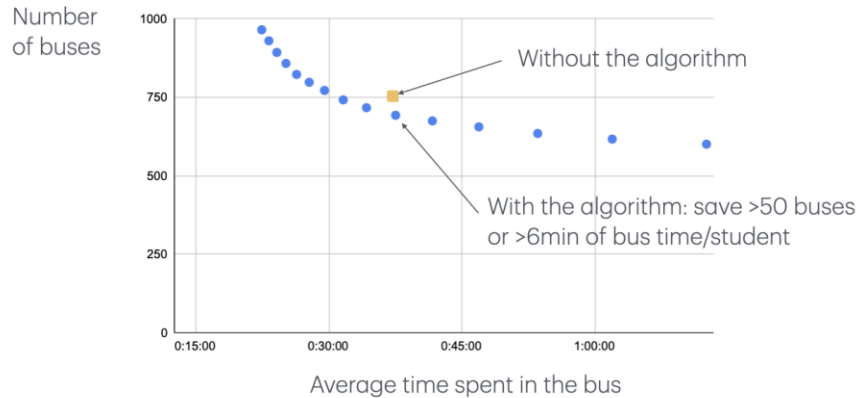
Then I mix in some background on the real Boston story so students see how the case maps onto a real situation. In the actual 2017 collaboration with MIT, Boston Public Schools already had a successful routing project under its belt: The algorithm pulled about 50 buses off the road and saved roughly \$5 million. *The Wall Street Journal* article “How Do You Fix a School Bus Problem? Call MIT” captures that phase well. This frames the optimizer for students not as a hypothetical but rather as something that has demonstrably worked.

We then walk through each character. For each one, I ask: “What did you learn from this person; what is their point of view; and why does it matter for your recommendation?” The class typically describes Sarah Lee’s optimism about the algorithm; Emily Carter’s balancing act across unions, mayoral politics, and community trust; James Thompson’s relentless focus on savings; Linda Johnson’s frontline skepticism about whether riders and/or drivers can absorb another change; and David Rodriguez’s reminders about student well-being. I emphasize the point that none of these voices is the villain; the district needs all five to get unstuck.

This is also the right moment to discuss the simulator and the trade-off it exposes. I project the Pareto curve of buses against average ride-time and walk students through it. (See **Figure TN3**.) The optimizer can cut dozens of buses with little change to mean ride-time, or it can keep the fleet size constant while shortening rides. Three lessons stand out: the optimizer makes a space of choices visible that a human planner could never enumerate; aggregate metrics hide individual

disruption (the same average can shift any specific student by 10 or more minutes); and choosing a point on the frontier is a governance decision, not a math problem.

Figure TN3: Pareto Frontier of Buses against Average Ride-Time



2. Harvesting and debating recommendations [~25 Minutes]

With a shared picture established, I ask each team to share what they recommended and why. Rather than chasing a single correct answer, we build a portfolio of moves: tuning the algorithm to cut buses; piloting ridership tracking because many buses run half-empty; cushioning union impact with driver supports; re-sequencing bell schedules for both equity and efficiency; or even targeted school consolidations.

I push teams to pressure-test each other. If a team proposes closing two schools, I ask the room which families would be displaced and how. If a team proposes algorithmic rerouting, I ask Linda Johnson’s question: How do you handle drivers who are already running marginal buses? When students get excited about a clean analytic answer, I press them to articulate what could go wrong: black-box mistrust, missing constraints, or equity impacts hiding under averages. A short slide summarizing the recommendations from the submitted memos can speed this up; otherwise, I let the conversation breathe.

If the room hasn’t surfaced bell-schedule changes by this point, students almost always raise the idea on their own. I let it sit on the table and signal that we will pick it up in the closing segment.

3. What happened in practice [~20 Minutes]

Here, as in many cases, the closing segment is what students remember most. I narrate the real Boston journey in two phases:

- **Phase one: routing.** The routing collaboration that opened this story succeeded. Planners took the algorithm-drafted routes and refined them with local judgment; the district then removed roughly 50 buses. This is the version of the story most consistent with classroom-style optimization: math finds candidates; humans choose and adjust.
- **Phase two: bell schedules.** *The Boston Globe* article “Computers Can Solve Your Problem. You May Not Like the Answer” captures the next chapter. After the routing win,

Boston Public Schools commissioned the same MIT team to redesign bell schedules across the district. The algorithm produced a plan that would shift the majority of high schoolers to later start times (consistent with adolescent sleep research); redistribute the most coveted morning slots more equitably; and further trim the transportation budget. The school committee approved the plan in December 2017 in front of a hopeful audience.

What followed surprised everyone. Roughly 85% of the district was about to receive new start times; hundreds of families were facing shifts as large as 9:30 a.m. to 7:15 a.m.; and the political momentum collapsed within two weeks. Parents from wealthier neighborhoods packed school committee meetings with “Families over Algorithms” signs; the NAACP and the Lawyers’ Committee for Civil Rights warned that the upheaval would hit lower-wage families hardest, regardless of equity intent; and the mayor pulled the plan. The superintendent eventually resigned in part because of the fallout.

I close by asking the class what this story teaches them. The lesson is not that the algorithm was wrong: it did exactly what it was asked to do, and the underlying optimization was sound. The lessons are that (1) operations decisions that look optimal on paper can collapse when they come into contact with politics and (2) change management, communication, and stakeholder sequencing are part of the analytical job—not a separate concern. As the MIT team’s advisor put it in the *Globe* piece: “But you put zero weight on the rest of society?”

Assessing Student Work

I grade the team memo on three things: the clarity of the recommendation; the quality of the evidence (do the cited conversations actually support the claim?); and the candor about trade-offs (does the team acknowledge what its plan would cost, and to whom?). I give bonus credit for creative levers students surfaced on their own, such as pilots for ridership tracking, driver-retention packages, or new data-collection ideas that address Linda Johnson’s concerns. Some instructors close the live session with a “mayor’s meeting” exercise, in which teams pitch their solutions while classmates role-play skeptical stakeholders who can force last-minute revisions.

Concluding Remarks

This case rewards instructors who let it run wide. Because it draws on a real, multi-pronged crisis, it can close a course on operations and analytics, but it also anchors conversations in adjacent areas. The angles below are the ones I have found most useful as final takeaways.

- **Prescriptive analytics in practice.** Optimization is not a single answer; it is a way of making a space of choices visible. The instructor’s job is not to find the optimum but to help students reason about the right point on the frontier given the values at stake.
- **Organization and stakeholder alignment.** Real operational change requires winning a coalition. Rodriguez’s equity worries, Johnson’s union realities, Thompson’s deficit math, and Carter’s political fragility are all genuine, and a good plan respects all of them.
- **Public-sector operations.** Public services carry constraints that private firms do not, such as equity mandates, public scrutiny, elected oversight, and unionized workforces. The case forces students to reckon with these constraints rather than abstract them away.

- **Why good ideas fail.** The Boston bell-schedule story is a study in how a technically sound, equity-improving plan can collapse when the people most affected feel it was done *to* them rather than *with* them. The lesson is not that the algorithm was wrong; it is that change management is part of analytics.
- **Consulting in practice.** Above all, the case is a simulation of consulting work: gathering information from people who do not always agree; deciding what to recommend; and defending it under questioning.

I hope the case works as beautifully in all classes as it has in mine. Instructors who would like to compare approaches or see additional materials are welcome to reach out to me directly at sebastien.martin@kellogg.northwestern.edu.